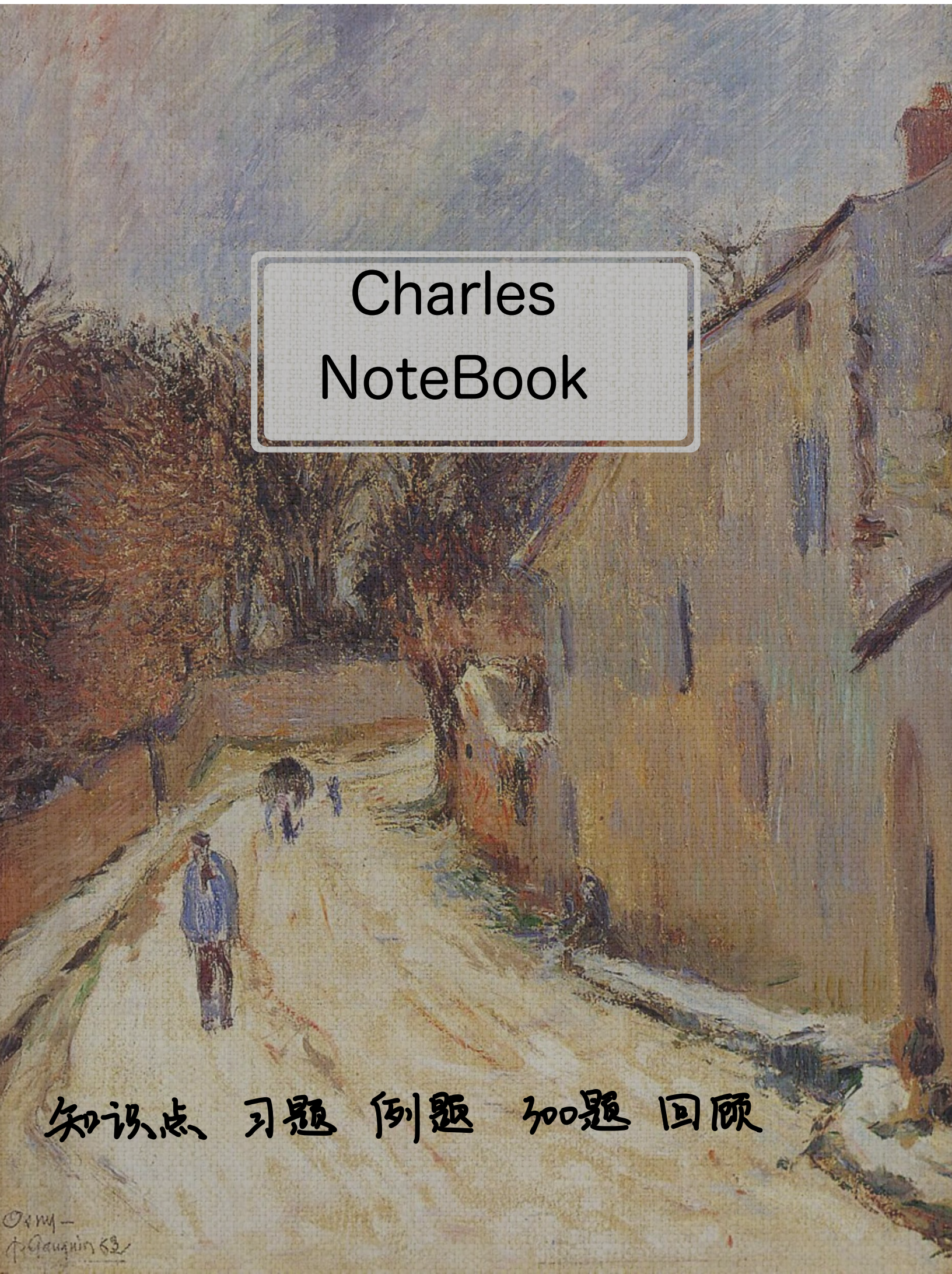


An impressionist painting of a street scene. In the foreground, a man in a blue coat and dark trousers stands on a light-colored path. To his right, a large, leafy tree with brown and orange foliage stands. In the background, more trees and a building with a red roof are visible. The sky is a mix of blue and white. The overall style is loose and painterly, with visible brushstrokes.

Charles NoteBook

知识点 习题 例题 加题 回顾

- 积分等式
 - 用中值定理
 - 用夹逼准则
 - 用积分法
- 积分不等式
 - 用函数单调性
 - 用拉氏
 - 用泰勒 ← 2022
 - 用积分法

常考: ① $\int_a^b$
 ② $\lim_{n \rightarrow \infty} \int_a^b$
 ③ $F(x) = \int_a^x f(t) dt$, $F(x)$ 单调性

例10.1 (背)

设 $f(x), g(x)$ 在 $[a, b]$ 上连续且 $g(x)$ 不变号,

证明 至少存在一点 $\xi \in [a, b]$ s.t. $\int_a^b f(x)g(x)dx = f(\xi) \int_a^b g(x)dx$

推广的积分中值定理

$$\text{令 } F(x) = \int_a^x f(t)g(t)dt$$

$$G(x) = \int_a^x g(t)dt$$

上限变量化

不妨设 $g(x) > 0$ 对 $F(x), G(x)$ 在 $[a, b]$ 上用柯西中值定理

$$\frac{F(b) - F(a)}{G(b) - G(a)} = \frac{f(\xi)}{g(\xi)} \Rightarrow \frac{\int_a^b f(x)g(x)dx}{\int_a^b g(x)dx} = \frac{F'(\xi)}{G'(\xi)} = \frac{f(\xi)g(\xi)}{g(\xi)} = f(\xi) \quad \xi \in (a, b) \subset [a, b]$$

$$\Rightarrow \int_a^b f(x)g(x)dx = f(\xi) \cdot \int_a^b g(x)dx$$

如(2023)

设 $f(x)$ 在 $[1, 2]$ 连续, 则计算 $\lim_{n \rightarrow \infty} \int_1^2 f(x) \cdot e^{-x^n} dx$

$f(x) \cdot g(x)$ 不变号

$$\lim_{n \rightarrow \infty} \int_1^2 f(x) e^{-x^n} dx \xrightarrow{\text{由积分中}} \lim_{n \rightarrow \infty} f(\xi_1) \cdot \int_1^2 e^{-x^n} dx$$

$$\lim_{n \rightarrow \infty} \int_1^2 e^{-x^n} dx \xrightarrow{\text{由积分中}} \lim_{n \rightarrow \infty} \frac{e^{-\xi_2^n}}{2-1} = \lim_{n \rightarrow \infty} e^{-\xi_2^n} = 0 \quad \xi_2 \in (1, 2)$$

$\Rightarrow$ 原=0

例10.2 ★

设函数 $f(x)$ 在 $[a, b]$ 上不恒为0, 且二阶可导, $f(a) > 0, f(b) > 0, \int_a^b f(x)dx = 0$

证明 (i) 存在不同的 $\xi_1, \xi_2 \in (a, b)$ s.t. $f(\xi_1) = f(\xi_2) = 0$

(ii) $f(x)$ 不恒为0, 假设 $f(x) \geq 0$ 与 $\int_a^b f(x)dx = 0$ 矛盾

$\therefore$ 存在 $f(\xi) < 0$

由零点定理 $f(a) > 0, f(b) > 0, f(\xi_1) = f(\xi_2) = 0$

证明 (ii) 存在 $\eta \in (a, b)$ 使得 $f''(\eta) > 0$

$$\frac{f(\xi_1) - f(a)}{\xi_1 - a} = f'(\eta_1) < 0$$

$$\frac{f(b) - f(\xi_2)}{b - \xi_2} = f'(\eta_2) > 0$$

$$\frac{f'(\eta_2) - f'(\eta_1)}{\eta_2 - \eta_1} = f''(\eta) > 0$$

$\eta_1 \in (a, \xi_1)$

$\eta_2 \in (\xi_2, b)$

$\eta \in (\eta_1, \eta_2)$

例10.3 夹逼

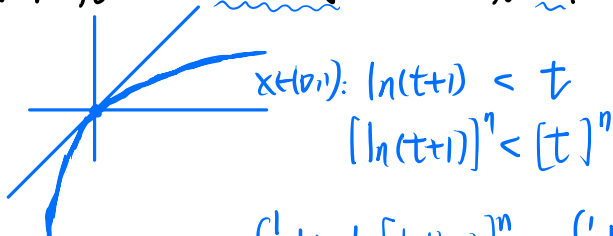
$$\lim_{n \rightarrow \infty} \int_0^1 \frac{x^n}{1+x} dx$$

$$0 \leq \frac{x^n}{1+x} \leq x^n$$

$$\int_0^1 x^n dx = \frac{x^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1} \rightarrow 0$$

例10.4

(1) 比较 $\int_0^1 |\ln t| \cdot [\ln(t+1)]^n dt$ 与 $\int_0^1 t^n |\ln t| dt$ ($n=1, 2, \dots$) 的大小, 说明理由



$$\int_0^1 |\ln t| \cdot [\ln(t+1)]^n dt < \int_0^1 |\ln t| \cdot t^n dt$$

(2) $u_n = \int_0^1 |\ln t| [\ln(t+1)]^n dt$ ($n=1, 2, \dots$) 求 $\lim_{n \rightarrow \infty} u_n$

$$\begin{aligned} & \int_0^1 |\ln t| t^n dt \\ &= - \int_0^1 \ln t \cdot t^n dt \\ &= \frac{1}{n+1} \int_0^1 \ln t \cdot d t^{n+1} \\ &= \frac{1}{n+1} \left[\ln t \cdot t^{n+1} \Big|_0^1 - \int_0^1 t^n dt \right] \\ &= \frac{1}{n+1} \int_0^1 t^n dt = \frac{1}{(n+1)^2} \end{aligned}$$

$$0 < \lim_{n \rightarrow \infty} u_n < \lim_{n \rightarrow \infty} \int_0^1 |\ln t| t^n dt = 0$$

由夹逼定理 $\lim_{n \rightarrow \infty} u_n = 0$

例10.5

设 $f(x)$ 是连续的偶函数, 且 T 周期函数

(1) 证明 $\int_0^{nT} x f(x) dx = \frac{n^2 T}{2} \int_0^T f(x) dx$ ($n=1, 2, 3, \dots$)

(2) 利用 (1) 计算 $I = \int_0^{n\pi} x |\sin x| dx$

$$\begin{aligned} (1) & \int_0^{nT} x f(x) dx \\ & \stackrel{\text{区间倒置}}{=} \int_0^{nT} (nT-x) f(nT-x) dx \\ &= \int_0^{nT} nT f(nT-x) dx - \int_0^{nT} x f(nT-x) dx \\ &= nT \int_0^{nT} f(x) dx - \int_0^{nT} x f(x) dx \\ \Rightarrow & \int_0^{nT} x f(x) dx = \frac{n^2 T}{2} \int_0^T f(x) dx \end{aligned}$$

$$\begin{aligned} (2) & I = \int_0^{n\pi} x |\sin x| dx \\ &= \frac{n^2 \pi}{2} \int_0^\pi |\sin x| dx \\ &= \frac{n^2 \pi}{2} \int_0^\pi \sin x dx \\ &= n^2 \pi \end{aligned}$$

例10.6 (单调性)

设 $f(x)$ 在 $[a, b]$ 上连续, 且 $f(x) > 0$,

证明: $\int_a^b f(x) dx \cdot \int_a^b \frac{1}{f(x)} dx \geq (b-a)^2$

$$\text{令 } F(x) = \int_a^x f(t) dt \cdot \int_a^x \frac{1}{f(t)} dt - (x-a)^2$$

$$F'(x) = f(x) \int_a^x \frac{1}{f(t)} dt + \frac{1}{f(x)} \int_a^x f(t) dt - 2(x-a)$$

$$= \int_a^x \frac{f(x)}{f(t)} dt + \int_a^x \frac{f(t)}{f(x)} dt - 2(x-a)$$

$$= \int_a^x \frac{f(x)}{f(t)} + \frac{f(t)}{f(x)} dt \geq 2 \int_a^x dt - 2(x-a) = 0$$

$\therefore F(x)$ 单调增加

$$\therefore F(b) \geq F(a) = 0 \quad *$$

例10.7

设函数 $f(x), g(x)$ 在 $[a, b]$ 连续, 且 $f(x)$ 单调增加,

$0 \leq g(x) \leq 1$. 证明:

$$(1) 0 \leq \int_a^x g(t) dt \leq x-a, \quad x \in [a, b]$$

$$(2) \int_a^{a+\int_a^x g(t) dt} f(u) du \leq \int_a^x f(t) g(t) dt$$

$$(1) 0 \leq g(x) \leq 1$$

$$\int_a^x 0 dt \leq \int_a^x g(t) dt \leq \int_a^x 1 dt$$

$$0 \leq \int_a^x g(t) dt \leq x-a$$

(2)

$$\text{令 } F(x) = \int_a^{a+\int_a^x g(t) dt} f(u) du - \int_a^x f(t) g(t) dt$$

$$F'(x) = f\left[a+\int_a^x g(t) dt\right] \cdot g(x) - f(x) \cdot g(x)$$

$$= g(x) \left[f\left(a+\int_a^x g(t) dt\right) - f(x) \right]$$

$$\therefore a+\int_a^x g(t) dt \leq x$$

由 $f(x) \uparrow$

$$\therefore F'(x) \geq 0$$

$$F(b) \geq F(a) = 0 \quad *$$

例10.8

设 $f(x)$ 在 $[0,1]$ 上具有一阶连续导数, 且 $f(0)=f(1)=0$.

证明 $\left| \int_0^1 f(x) dx \right| \leq \frac{1}{4} \max_{x \in [0,1]} |f'(x)|$

见到 $f, f' \Rightarrow$ 拉氏

- ① 区间
- ② 对象
- ③ 用定理
- ④ 条件

$$\begin{aligned} (1) \quad [0, x] &\Rightarrow f(x) - f(0) = f'(\xi_1) \cdot x \quad \xi_1 \in (0, x) \\ [x, 1] &\Rightarrow f(1) - f(x) = f'(\xi_2) \cdot (1-x) \quad \xi_2 \in (x, 1) \\ \Rightarrow \begin{cases} |f(x)| = |f'(\xi_1) x| = |f'(\xi_1)| x \\ |f(x)| = |f'(\xi_2) (1-x)| = |f'(\xi_2)| (1-x) \end{cases} \end{aligned}$$

$$\left| \int_0^1 f(x) dx \right| \leq \int_0^1 |f(x)| dx$$

(2) 一阶连续 $\Rightarrow$ 有界. $|f(x)| \leq M = \max_{x \in [0,1]} |f'(x)|$

$$\Rightarrow \begin{cases} |f'(\xi_1)| \cdot x \leq M \cdot x \\ |f'(\xi_2)| (1-x) \leq M (1-x) \end{cases}$$

$$(3) \quad \left| \int_0^1 f(x) dx \right| = \left| \int_0^x f(t) dt + \int_x^1 f(t) dt \right| \leq \left| \int_0^x f(t) dt \right| + \left| \int_x^1 f(t) dt \right|$$

$$\leq \int_0^x |f(t)| dt + \int_x^1 |f(t)| dt \leq \int_0^x M t dt + \int_x^1 M (1-t) dt = M \cdot \frac{x^2}{2} + M \cdot \frac{(1-x)^2}{2}$$

$$= M \cdot \left[\frac{x^2}{2} + \frac{(1-x)^2}{2} \right] = \max_{x \in [0,1]} |f'(x)| \cdot \left[\frac{x^2}{2} + \frac{(1-x)^2}{2} \right]$$

$$\min \left[\frac{x^2}{2} + \frac{(1-x)^2}{2} \right] = \frac{1}{4}$$

例10.9

设 $f(x)$ 在 $[0,2]$ 上二阶可导, 且 $f(1)=0$, 当 $x \in [0,2]$ 时, 记 $M = \max_{x \in [0,2]} |f''(x)|$, 证明 $\left| \int_0^2 f(x) dx \right| \leq \frac{1}{3} M$

$$\begin{cases} \text{拉} & f(x) - f(x_0) = f'(x_0)(x-x_0) \\ \text{泰} & f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(\xi)}{2} (x-x_0)^2 \end{cases}$$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(\xi)}{2} (x-x_0)^2$$

$$f(x) \leq f(x_0) + f'(x_0)(x-x_0) + \frac{M}{2} (x-x_0)^2$$

$$f(x) \leq f(1) + f'(1)(x-1) + \frac{M}{2} (x-1)^2 = f'(1)(x-1) + \frac{M}{2} (x-1)^2$$

$$\left| \int_0^2 f(x) dx \right| \leq \int_0^2 |f(x)| dx = \frac{M}{2} \int_0^2 (x-1)^2 dx = \frac{M}{3}$$

例10.10 ~~★★★~~

设 $f(x)$ 的 n -阶导数 $f^{(n)}(x)$ 在 $[0,1]$ 上连续.

且 $f(0)=f(1)=0$, 证明

$$(1) \int_0^1 f(x) dx = \frac{1}{2} \int_0^1 x(x-1) f^{(n)}(x) dx$$

$$(2) \left| \int_0^1 f(x) dx \right| \leq \frac{1}{12} \max_{0 \leq x \leq 1} \{ |f^{(n)}(x)| \}$$

$$\begin{aligned} (1) & \int_0^1 \frac{x(x-1)}{2} f^{(n)}(x) dx \\ &= \int_0^1 \frac{x^2-x}{2} d f^{(n-1)}(x) \\ &= \frac{x^2-x}{2} f^{(n-1)}(x) \Big|_0^1 - \frac{1}{2} \int_0^1 f^{(n-1)}(x) (2x-1) dx \\ &= -\frac{1}{2} \int_0^1 (2x-1) d f^{(n-2)}(x) \\ &= -\left[(2x-1) f^{(n-2)}(x) \Big|_0^1 - \int_0^1 f^{(n-2)}(x) d(2x-1) \right] \\ &= \int_0^1 f^{(n-2)}(x) dx \end{aligned}$$

$$\begin{aligned} (2) \left| \int_0^1 f(x) dx \right| &= \frac{1}{2} \left| \int_0^1 x(x-1) f^{(n)}(x) dx \right| \\ &\leq \frac{M}{2} \int_0^1 x(x-1) dx = \frac{M}{12} \end{aligned}$$

用中值定理 $g(x)$ 不变号, $\int_a^b f(x)g(x) dx = f(\eta) \int_a^b g(x) dx$ 所使用与证明

用来逼收敛

用积分法 通常分两步, 第一步求积分等于, 第二步用第一步结果求一较难积分

用单调性 把不等式移到一边, 上限变量化, 求导.

用拉格朗日 $f(x) - f(x_0) = f'(\eta)(x - x_0)$

用泰勒 (2022) 高阶可导, 某端点, 简单 $f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(\eta)}{2}(x-x_0)^2$

用积分法

习10.1

设 $f(x), g(x)$ 在 $[a, b]$ 上连续, 证明至少存在一点 $\xi \in (a, b)$, 使得

$$f(\xi) \int_{\xi}^b g(x) dx = g(\xi) \int_a^{\xi} f(x) dx$$

$$\text{证 } f(\xi) \int_{\xi}^b g(x) dx - g(\xi) \int_a^{\xi} f(x) dx = 0$$

$$F(x) = \int_a^x f(x) dx \int_x^b g(x) dx \quad \text{证 } F'(\xi) = 0$$

$$F(a) = F(b) = 0 \Rightarrow F'(\xi) = 0$$

习10.2

设 $\varphi(x)$ 是可微函数 $f(x)$ 的反函数, 且 $f(1) = 0$.

$$\text{证明 } \int_0^1 \left[\int_0^{f(x)} \varphi(t) dt \right] dx = 2 \int_0^1 x f(x) dx$$

$$F(x) = \int_0^{f(x)} \varphi(t) dt$$

$$F'(x) = \varphi(f(x)) \cdot f'(x) = x \cdot f'(x)$$

$$\tau_1 = \int_0^1 F(x) dx = \int_0^1 x f'(x) dx = \int_0^1 x d f(x) = x f(x) \Big|_0^1 - \int_0^1 f(x) dx$$

$$= - \int_0^1 f(x) dx$$

$$\tau_2 = 2 \int_0^1 x f(x) dx = \int_0^1 f(x) d x^2 = x^2 f(x) \Big|_0^1 - \int_0^1 2x f(x) dx = - \int_0^1 f(x) dx$$

$$\Rightarrow \tau_1 = \tau_2$$

习10.3

设 $f(x)$ 在 $[a, b]$ 上连续且严格单调增加, 证明

$$(a+b) \int_a^b f(x) dx < 2 \int_a^b x f(x) dx$$

$$F(x) = (a+x) \int_a^x f(t) dt - 2 \int_a^x t f(t) dt \quad \text{证 } F(b) < 0$$

$$F'(x) = \int_a^x f(t) dt + (a+x) \cdot f(x) - 2x f(x)$$

$$= \int_a^x f(t) dt - (x-a) f(x)$$

$$= (x-a)(f(\xi) - f(x)) \quad \xi \in (a, x)$$

$$\because f(x) \text{ 单调递增}$$

$$\therefore F'(x) < 0 \quad \therefore F(x) \text{ 单调减.}$$

$$F(a) = 0 \quad F(x) \downarrow$$

$$\therefore F(b) < 0. \quad *$$

习10.4

设 $f(x)$ 在 $[0, 1]$ 上连续且单调减少,

证明: 当 $0 < \lambda < 1$ 时, $\int_0^\lambda f(x) dx \geq \lambda \int_0^1 f(x) dx$

$$\text{证 } \frac{\int_0^\lambda f(x) dx}{\lambda} \geq \int_0^1 f(x) dx$$

$$F(x) = \frac{\int_0^x f(t) dt}{x}$$

$$\begin{aligned} F'(x) &= \frac{x f(x) - \int_0^x f(t) dt}{x^2} = \frac{x f(x) - \int_0^x f(t) dt}{x^2} \\ &= \frac{x f(x) - x f(\xi)}{x^2} = \frac{f(x) - f(\xi)}{x} \quad \xi \in (0, x) \end{aligned}$$

$\therefore f(x)$ 单调减少

$\therefore F'(x) < 0$

$\therefore F(x) < F(1) \quad \times$

习10.5

$$\text{证明 } \int_0^{\frac{\pi}{2}} \frac{\cos x}{1+x^2} dx \geq \int_0^{\frac{\pi}{2}} \frac{\sin x}{1+x^2} dx \quad x - \frac{\pi}{2} = t$$

$$\Rightarrow \int_0^{\frac{\pi}{4}} \frac{\cos x - \sin x}{1+x^2} dx \geq \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{\sin x - \cos x}{1+x^2} dx$$

$$\Rightarrow \int_0^{\frac{\pi}{4}} \frac{\cos x - \sin x}{1+x^2} dx \geq \int_0^{\frac{\pi}{4}} \frac{\sin(x+\frac{\pi}{4}) - \cos(x+\frac{\pi}{4})}{1+(x+\frac{\pi}{4})^2} dt = \int_0^{\frac{\pi}{4}} \frac{\sqrt{2} \sin x}{1+(x+\frac{\pi}{4})^2} dx = \int_0^{\frac{\pi}{4}} \frac{\sqrt{2} \sin(\frac{\pi}{4}-x)}{1+(\frac{\pi}{2}-x)^2} dx$$

$$\Rightarrow \int_0^{\frac{\pi}{4}} \frac{\sqrt{2} \sin(\frac{\pi}{4}-x)}{1+x^2} dx \geq \int_0^{\frac{\pi}{4}} \frac{\sqrt{2} \sin(\frac{\pi}{4}-x)}{1+(\frac{\pi}{2}-x)^2} dx$$

$$\begin{aligned} F(x) &= (1+x^2) - (1+(\frac{\pi}{2}-x)^2) \\ &= x^2 - (\frac{\pi}{2}-x)^2 = x^2 - \frac{\pi^2}{4} - x^2 + \pi x = x\pi - \frac{\pi^2}{4} \end{aligned}$$

当 $x \in (0, \frac{\pi}{4})$, $F(x) \leq 0$

$\therefore$ 当 $x \in (0, \frac{\pi}{4})$, $0 < 1+x^2 \leq 1+(\frac{\pi}{2}-x)^2 \quad \times$

300.10.1

设 $f(x)$ 在 $[0, \pi]$ 连续, 证明

$$(1) \int_0^{\pi/2} f(\sin x) dx = \int_0^{\pi/2} f(\cos x) dx$$

$$\int_0^{\pi/2} f(\sin x) dx \stackrel{\text{区间再现}}{=} \int_0^{\pi/2} f(\sin(\frac{\pi}{2}-x)) dx = \int_0^{\pi/2} f(\cos x) dx$$

$$(2) \int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$$

$$\int_0^{\pi} x f(\sin x) dx = \int_0^{\pi} (\pi-x) f(\sin(\pi-x)) dx = \int_0^{\pi} (\pi-x) f(\sin x) dx$$

$$= \pi \int_0^{\pi} f(\sin x) dx - \int_0^{\pi} x f(\sin x) dx$$

$$\Rightarrow \int_0^{\pi} x f(\sin x) dx = \frac{\pi}{2} \int_0^{\pi} f(\sin x) dx$$

300.10.2

设 $f(x)$ 在 $[-\pi, \pi]$ 连续, 且有 $f(x) = \frac{x}{1+\cos^2 x} + \int_{-\pi}^{\pi} f(x) \sin x dx$ 求 $f(x)$

$$f(x) = \frac{x}{1+\cos^2 x} + A$$

$$\int_{-\pi}^{\pi} f(x) \sin x dx = A = \int_{-\pi}^{\pi} \frac{x \sin x}{1+\cos^2 x} + A \sin x dx$$

$$A = \int_{-\pi}^{\pi} \frac{x \sin x}{1+\cos^2 x} dx = 2 \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} = 2 \int_0^{\pi} \frac{x d \cos x}{1+\cos^2 x} \stackrel{t=\cos x}{=} 2 \int_1^{-1} \frac{\arccos x}{1+x^2} dx$$

$$= 2 \int_1^{-1} \arccos x d \arctan x = 2 \arccos x \cdot \arctan x \Big|_1^{-1} + 2 \int_1^{-1} \frac{\arctan x}{1+x^2} dx$$

$$= \underbrace{2(\pi \cdot \frac{3\pi}{2} - 0 \cdot \frac{\pi}{2})}_{3\pi^2} + 2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{t \cdot \sec^2 t}{\sec t} dt$$

$$x = \tan t$$

$$= 3\pi^2 + 2 \int_{-\pi/2}^{\pi/2} t \sec t dt = 3\pi^2$$

$$f(x) = \frac{x}{1+\cos^2 x} + 3\pi^2$$

300.10.3

设 $\varphi(x)$ 是可微函数 $f(x)$ 的反函数, $f(0)=0$

证明 $\int_0^1 \left[\int_0^{f(x)} \varphi(t) dt \right] dx = 2 \int_0^1 x f(x) dx$

$$\int_a^b f(x) dx = (b-a)f(\eta)$$

$$\begin{aligned} & \text{左} \int_0^1 \left[\int_0^{f(x)} \varphi(t) dt \right] dx \\ &= \left[\int_0^{f(x)} \varphi(t) dt \right] \cdot x \Big|_0^1 - \int_0^1 x d \left[\int_0^{f(x)} \varphi(t) dt \right] \\ &= \underbrace{\int_0^{f(1)} \varphi(t) dt}_0 - \int_0^1 x \cdot f'(x) \cdot \varphi[f(x)] \cdot dx \\ &= - \int_0^1 x^2 f'(x) dx \end{aligned}$$

$$\text{右} = 2 \int_0^1 x f(x) dx = \int_0^1 f(x) dx^2 = \underbrace{x^2 f(x)}_0 \Big|_0^1 - \int_0^1 x^2 f'(x) dx = - \int_0^1 x^2 f'(x) dx$$

左=右 *

300.10.4

设 $f(x)$ 在 $[0,1]$ 连续, $(0,1)$ 可导,

且 $\int_{2/3}^1 f(x) dx = f(0)$ 证明: 在 $(0,1)$ 内存在 c , $f'(c)=0$.

$$\int_{2/3}^1 f(x) dx = (1 - \frac{2}{3}) \cdot f(\eta) = f(\eta), \quad \eta \in (0,1)$$

$$f(\eta) = f(0) \Rightarrow f'(c) = 0, \quad c \in (0, \eta) \subset (0,1)$$

300.10.5

(1) 证明积分中值: $f(x), [a,b]$ 连续, $\eta \in [a,b] \rightarrow \int_a^b f(x) dx = f(\eta)(b-a)$

$$\text{设 } F(x) = \int_a^x f(t) dt$$

$$\int_a^b f(x) dx = F(b) - F(a) = f(\eta)(b-a) ?$$

(2) 若函数 $\varphi(x)$ 具有二阶导数, 且满足 $\varphi(2) > \varphi(1)$, $\varphi(2) > \int_1^2 \varphi(x) dx$
至少存在一点 $\xi \in (1,2)$, 使得 $\varphi''(\xi) < 0$

$$\begin{aligned} & \Rightarrow \varphi'(\xi_1) > 0 \quad \varphi'(\xi_2) < 0 \\ \text{① } \varphi_2 > \varphi_1 & \Rightarrow \varphi_2 - \varphi_1 = (2-1) \cdot \varphi'(\xi_1) > 0 \Rightarrow \varphi'(\xi_1) > 0 \quad \xi_1 \in (0,1) \end{aligned}$$

$$\textcircled{2} \quad \varphi(2) > \int_2^3 \varphi(x) dx = (3-2) \varphi(\eta) = \varphi(\eta), \quad \eta \in (2,3) \\ \Rightarrow \varphi(2) - \varphi(\eta) = (2-\eta) \varphi'(\xi_2) > 0 \Rightarrow \varphi'(\xi_2) < 0, \quad \xi_2 \in (2, \eta)$$

$$\textcircled{1} \text{ 由零点定理 } \Rightarrow \varphi'(\xi) < 0, \quad \xi \in (\xi_1, \xi_2) \subset (1,3)$$

300.10.6

设 $f(x), g(x)$ 的导数在 $[0,1]$ 上连续, 且 $f(0)=0, f'(x) \geq 0, g'(x) \geq 0$,
证明 $\forall a \in [0,1]$, 有 $\int_0^a g(x) f'(x) dx + \int_0^1 f(x) g'(x) dx \geq f(a) g(1)$

$$\begin{aligned} F(x) &= \int_0^x g(x) df(x) + \int_0^1 f(x) dg(x) - f(x)g(1) \\ &= g(x)f(x) \Big|_0^x - \int_0^x f(x) dg(x) + \int_0^1 f(x) dg(x) - f(x)g(1) \\ &= f(x)[g(x)-g(1)] + \int_a^1 f(x) g'(x) dx \\ &= -f(x) \int_a^1 g'(x) dx + \int_a^1 f(x) g'(x) dx \\ &= \int_a^1 [f(x) - f(a)] g'(x) dx \end{aligned}$$

$$\begin{aligned} f(x) \nearrow \quad & x \in (a,1) \text{ 时 } f(x) - f(a) > 0, \\ & g'(x) \geq 0 \\ \Rightarrow F(x) & \geq 0 \end{aligned}$$

300.10.7

设 $f'(x)$ 在 $[0,a]$ 上连续, $f(0)=0$

$$\text{证明: } \left| \int_0^a f(x) dx \right| \leq \frac{1}{2} M a^2, \quad M = \max_{0 \leq x \leq a} \{ |f'(x)| \}$$

$$\begin{aligned} \text{泰勒: } f(x) &= f(x_0) + f'(\xi)(x-x_0) \\ x_0=0 \text{ 时 } f(x) &= f'(\xi)x \end{aligned}$$

$$\left| \int_0^a f(x) dx \right| \leq \int_0^a |f(x)| dx = \int_0^a |f'(\xi)x| dx = M \int_0^a x dx = \frac{M}{2} a^2$$

300.10.8

设 $f'(x)$ 在 $[a,b]$ 连续, 且 $f(a)=f(b)=0$.

$$\text{证明 } |f(x)| \leq \frac{1}{2} \int_a^b |f'(x)| dx$$

$$f(x) = f(x) - f(a) = \int_a^x f'(t) dt$$

$$|f(x)| \leq \int_a^x |f'(t)| dt \quad \textcircled{1}$$

$$f(x) = f(x) - f(b) = \int_b^x f'(t) dt$$

$$|f(x)| \leq \int_x^b |f'(t)| dt \quad \textcircled{2}$$

$$\textcircled{1} + \textcircled{2} \quad |f(x)| \leq \frac{1}{2} \int_a^b |f'(x)| dx$$

300.10.9

$$\int_a^x f(t) dt \geq \int_a^x g(t) dt, \quad x \in [a, b).$$

$$\int_a^b f(t) dt = \int_a^b g(t) dt.$$

$$\text{ii} \Rightarrow \int_a^b x f(x) dx \leq \int_a^b x g(x) dx$$

$$Z(x) = \int_a^x f(t) dt - \int_a^x g(t) dt = \int_a^x f(t) - g(t) dt \geq 0$$

$$Z(a) = Z(b) = 0$$

$$\text{iii} \quad W(x) = \int_a^x t f(t) - t g(t) dt \quad W(b) \leq 0.$$

$$= \int_a^x t [f(t) - g(t)] dt$$

$$= \int_a^x t d[F(t) - G(t)]$$

$$= t[F(t) - G(t)] \Big|_a^b - \int_a^b F(t) - G(t) dt$$

$$= - \int_a^b Z(t) dt$$

$$\therefore Z(t) \geq 0$$

$$\therefore W(b) \leq 0 \quad \times$$

300.10.10

设 $f(x)$ 在 $[0,1]$ 二阶可导, $f(\frac{1}{2})=1$ $f''(x)>0$

证明 $\int_0^1 f(x) dx \geq 1$

$$\begin{aligned}\int_0^1 f(x) dx &= \int_0^1 \underbrace{f(\frac{1}{2})}_{1} + \underbrace{f'(\frac{1}{2})(x-\frac{1}{2})}_{0} + f''(\xi)(x-\frac{1}{2})^2 dx \\ &= 1 + 0 + \int_0^1 f''(\xi)(x-\frac{1}{2})^2 dx \geq 1\end{aligned}$$